Worksheet 3.5 — The Unit Circle and Trigonometric Identities

Mathematics: Analysis & Approaches SL (IB) · Unit 3: Geometry and Trigonometry

The Big Ideas

The unit circle and the Pythagorean identity let you find exact trig values in any quadrant.

- $\sin^2\theta + \cos^2\theta = 1$, $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.
- Quadrant signs (All-Students-Take-Calculus): Q1 all +, Q2 sin +, Q3 tan +, Q4 cos +.

Worked Examples

Example 1 — Exact value

Find the exact value of $\sin(300^\circ)$.

Solution. Reference angle 60°, Q4 (sin negative): $-\sin(60^\circ) = -\dfrac{\sqrt{3}}{2}$.

Conclusion: $-\dfrac{\sqrt{3}}{2}$.

Example 2 — Quadrant and Pythagorean identity

Given $\sin\theta = -\dfrac{5}{13}$, θ in Q3, find $\cos\theta$.

Solution. $\cos^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$; Q3 cosine negative.

Conclusion: $\cos\theta = -\dfrac{12}{13}$.

Example 3 — Prove an identity

Prove $(\sin x + \cos x)^2 = 1 + 2\sin x\cos x$.

Solution. Expand: $\sin^2 x + 2\sin x\cos x + \cos^2 x = 1 + 2\sin x\cos x$ (using the Pythagorean identity).

Conclusion: identity confirmed.

Example 4 — From tangent to sine and cosine

Given $\tan\theta = -\dfrac{3}{4}$, θ in Q2, find $\sin\theta$ and $\cos\theta$.

Solution. Reference triangle: opp 3, adj 4, hyp 5. Q2: sin +, cos −.

Conclusion: $\sin\theta = \dfrac{3}{5}$, $\cos\theta = -\dfrac{4}{5}$.

Example 5 — Exact value in Q3

Find the exact value of $\tan(210ř)$.

Solution. Reference angle 30ř, Q3 (tan positive): $\tan(30ř) = \dfrac{1}{\sqrt{3}}$.

Conclusion: $\tan(210ř) = \dfrac{\sqrt{3}}{3}$.

Example 6 — Harder identity proof — factoring a difference of squares

Prove $\sin^4 x - \cos^4 x = \sin^2 x - \cos^2 x$.

Solution. Factor the left side as a difference of squares: $(\sin^2 x - \cos^2 x)(\sin^2 x + \cos^2 x)$.

Conclusion: the second factor is 1 by the Pythagorean identity, leaving $\sin^2 x - \cos^2 x$ — identity confirmed.

Example 7 — Determining a quadrant from two sign conditions

Given $\sin\theta < 0$ and $\cos\theta < 0$, determine the quadrant of θ.

Solution. Check each quadrant's signs: only Q3 has both sine and cosine negative.

Conclusion: θ is in Quadrant III.

Example 8 — Exact values from a given cosine

Given $\cos\theta = -\dfrac{2}{3}$, θ in Q2, find $\sin\theta$ and $\tan\theta$ exactly.

Solution. $\sin^2\theta = 1 - \frac{4}{9} = \frac{5}{9} \Rightarrow \sin\theta = \dfrac{\sqrt{5}}{3}$ (positive in Q2).

Conclusion: $\sin\theta = \dfrac{\sqrt{5}}{3}$, $\tan\theta = \dfrac{\sqrt{5}/3}{-2/3} = -\dfrac{\sqrt{5}}{2}$.

Example 9 — Deriving a secant-form identity

Prove that $1 + \dfrac{\sin^2 x}{\cos^2 x} = \dfrac{1}{\cos^2 x}$.

Solution. Combine the left side over a common denominator: $\dfrac{\cos^2 x + \sin^2 x}{\cos^2 x}$.

Conclusion: the numerator is 1 by the Pythagorean identity, giving $\dfrac{1}{\cos^2 x}$ — identity confirmed.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find the exact value of $\cos(150\check{r})$.

Question 2

Given $\cos\theta = \dfrac{8}{17}$, θ in Q4, find $\sin\theta$.

Question 3

Prove that $\tan x \sin x + \cos x = \dfrac{1}{\cos x}$.

Question 4 — Challenge

Prove that $1 - \dfrac{\sin^2 x}{1+\cos x} = \cos x$.

Question 5

Given $\tan\theta = 2$, θ in Q3, find $\sin\theta + \cos\theta$ exactly.

Question 6

Find the exact value of $\tan(225ř)$.

Question 7

Given $\sin\theta = 0.6$ and θ in Q2, find $\tan\theta$ exactly.

Question 8

Find the exact value of $\cot(240ř)$ (i.e. $1/\tan(240ř)$).

Question 9

Given $\sin\theta = -\frac{3}{5}$, θ in Q4, find $\cos\theta$ and $\tan\theta$ exactly.

Question 10

Prove that $\cos^4 x - \sin^4 x = 2\cos^2 x - 1$.

Question 11 — Challenge

Prove that $\dfrac{\cos x}{1+\sin x} = \dfrac{1-\sin x}{\cos x}$.

Question 12 — Challenge

Given $\tan\theta = -\dfrac{5}{12}$, θ in Q2, find $\sin\theta + \cos\theta$ exactly.

Question 13 — Challenge

Given $\tan\theta > 0$ and $\sin\theta < 0$, determine the quadrant of θ, explaining your reasoning.

Question Recap & Answer Key

Question Recap

1. Find the exact value of $\cos(150\text{ř})$.
2. Given $\cos\theta = \frac{8}{17}$, θ in Q4, find $\sin\theta$.
3. Prove that $\tan x \sin x + \cos x = \frac{1}{\cos x}$.
4. Prove that $1 - \frac{\sin^2 x}{1+\cos x} = \cos x$.
5. Given $\tan\theta = 2$, θ in Q3, find $\sin\theta + \cos\theta$ exactly.
6. Find the exact value of $\tan(225\text{ř})$.
7. Given $\sin\theta = 0.6$ and θ in Q2, find $\tan\theta$ exactly.
8. Find the exact value of $\cot(240\text{ř})$ (i.e. $1/\tan(240\text{ř})$).
9. Given $\sin\theta = -\frac{3}{5}$, θ in Q4, find $\cos\theta$ and $\tan\theta$ exactly.
10. Prove that $\cos^4 x - \sin^4 x = 2\cos^2 x - 1$.
11. Prove that $\frac{\cos x}{1+\sin x} = \frac{1-\sin x}{\cos x}$.
12. Given $\tan\theta = -\frac{5}{12}$, θ in Q2, find $\sin\theta + \cos\theta$ exactly.
13. Given $\tan\theta > 0$ and $\sin\theta < 0$, determine the quadrant of θ, explaining your reasoning.

✓ Answer Key

1. $-\frac{\sqrt{3}}{2}$
2. $\sin^2\theta = 1 - 64/289 = 225/289 \Rightarrow \sin\theta = -15/17$ (Q4)
3. $\tan x \sin x + \cos x = \frac{\sin^2 x}{\cos x} + \cos x = \frac{\sin^2 x + \cos^2 x}{\cos x} = \frac{1}{\cos x}$
4. $1 - \frac{1-\cos^2 x}{1+\cos x} = 1 - \frac{(1-\cos x)(1+\cos x)}{1+\cos x} = 1 - (1 - \cos x) = \cos x$
5. reference triangle opp 2 adj 1 hyp $\sqrt{5}$; Q3 both negative: $\sin\theta = -2/\sqrt{5}$, $\cos\theta = -1/\sqrt{5}$; sum $= -3/\sqrt{5} = -3\sqrt{5}/5$
6. 225ř is Q3, reference 45ř: $\tan(225\text{ř}) = 1$
7. $\cos\theta = -0.8$ (Q2 negative); $\tan\theta = 0.6/(-0.8) = -3/4$
8. 240ř is Q3, reference 60ř: $\tan(240\text{ř}) = \sqrt{3}$, so $\cot(240\text{ř}) = 1/\sqrt{3} = \sqrt{3}/3$
9. $\cos^2\theta = 1 - 9/25 = 16/25 \Rightarrow \cos\theta = 4/5$ (Q4 positive); $\tan\theta = -3/4$
10. cross-multiply: $\cos^2 x = (1 - \sin x)(1 + \sin x) = 1 - \sin^2 x$, true by the Pythagorean identity
11. factor as $(\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) = \cos^2 x - \sin^2 x = \cos^2 x - (1 - \cos^2 x) = 2\cos^2 x - 1$
12. reference triangle opp 5 adj 12 hyp 13; Q2: sin +, cos −: $\sin\theta = 5/13$, $\cos\theta = -12/13$; sum $= -7/13$
13. $\tan\theta > 0$ needs sin,cos same sign; combined with $\sin\theta < 0$, both must be negative: Quadrant III